Answer **all** the questions in the spaces provided.

1 A small block is released from rest on a frictionless surface inclined at 35° to the horizontal and slides a certain distance before reaching points A and B, as shown in Fig. 1.1.

The block was released at time $t = 0$, passing point A at time t_1 and point B at time t_2.

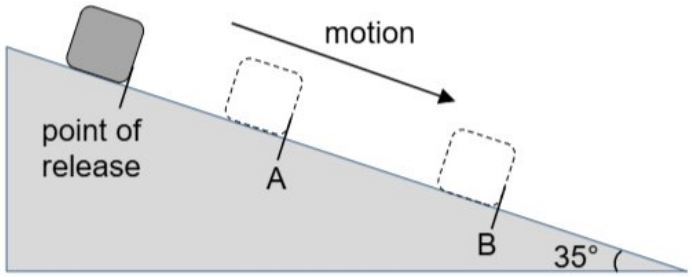


Fig. 1.1

(a) (i) Show that the acceleration of the block sliding down the frictionless surface is 5.6 m s^{-2}.

[1]

(ii) Sketch, in Fig. 1.2, the variation of the block's velocity v with time t, from $t = 0$ until the block reaches B. Mark the times t_1 and t_2 on the graph. No calculations are required.

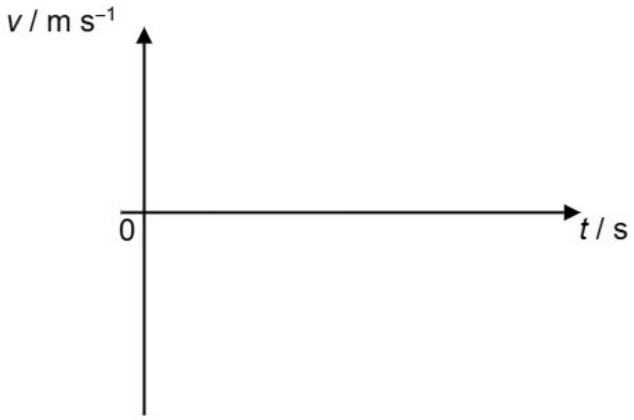


Fig. 1.2

[1]

(iii) The block takes 0.30 s to travel from A to B and reaches B with a velocity of 2.0 m s^{-1}. Hence or otherwise, determine the distance between A and B.

distance = ... m [2]

(iv) With reference to the definition of power, explain why the rate of increase of the block's kinetic energy with respect to time increases as the block moves down the incline.

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..... [2]

(b) The block is now fitted with a large piece of cardboard, with its surface facing the direction of motion, and is released from rest at the same position on the frictionless 35° incline. It accelerates down the slope and eventually reaches terminal velocity due to air resistance.

(i) Explain why the block eventually reaches terminal velocity.

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..... [2]

(ii) Sketch, in Fig. 1.3, a graph showing the variation of the block's acceleration *a* with time *t* for the block fitted with the piece of cardboard.

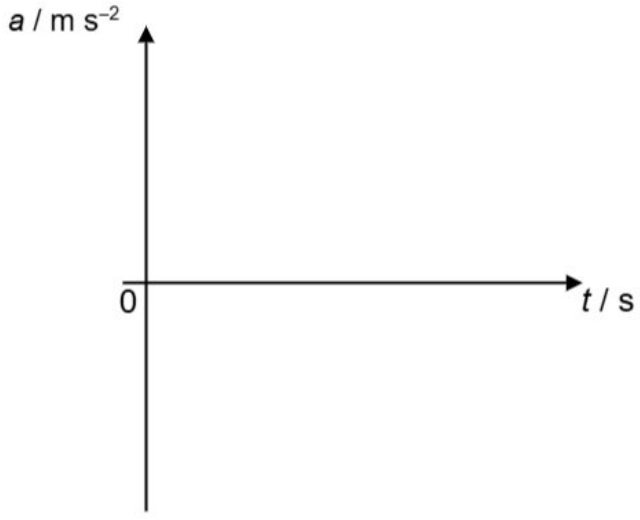


Fig. 1.3

[1]

(iii) Sketch, in Fig. 1.4, a graph showing the variation of the block's displacement *s* with time *t* for the block fitted with the piece of cardboard.

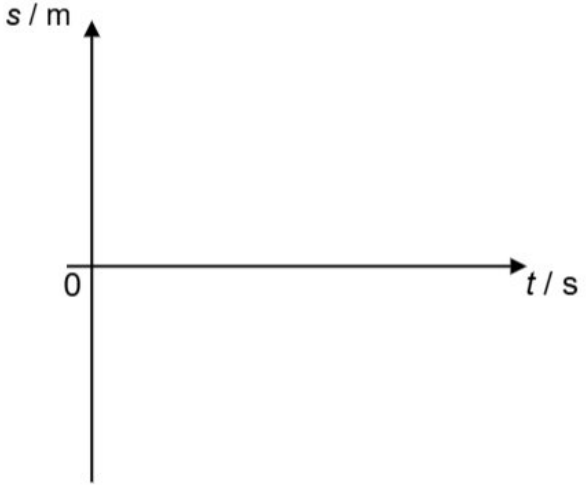


Fig. 1.4

[1]